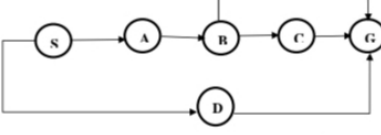
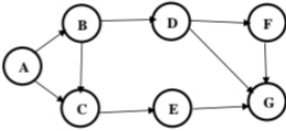
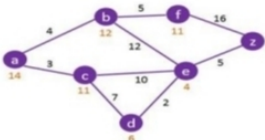


model question paper

Q. No.	Questions	Marks	BL
Module 1			
1	a. Define Artificial Intelligence. Explain the foundations of AI in detail.	10	CL2
	b. Discuss the PEAS specification of Biometric Authentication System	10	CL3
OR			
2	Differentiate: i. Fully observable vs partially observable ii. Single agent vs Multiagent iii. Deterministic vs Stochastic iv. Static vs Dynamic	10	CL2
	b. Give PEAS specification for Automated Taxi Driver	10	CL3
Module 2			
3	a. Explain five components and well-defined problem. Consider an 8-puzzle problem as an example.	10	CL2
	b. In detail elaborate with neat diagram, the state space for the vacuum world. Links denote actions: L = Left, R = Right, S = Suck.	10	CL3
OR			
4	a. For the graph given below apply BFS and DFS Search algorithm. 	10	CL3
	b. Explain Depth Limited Search and Iterative Deepening DFS Search with an example.	10	CL2
Module 3			
5	a. In the below graph, find the path from A to G. Using Greedy Best First search and A* search algorithm. The values in the table represent heuristic values of reaching the goal node G pass current node.	10	CL3

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	 <table border="1" data-bbox="979 925 1053 1048"> <tr><td>A</td><td>5</td></tr> <tr><td>B</td><td>6</td></tr> <tr><td>C</td><td>4</td></tr> <tr><td>D</td><td>3</td></tr> <tr><td>E</td><td>3</td></tr> <tr><td>F</td><td>1</td></tr> <tr><td>G</td><td>0</td></tr> </table>	A	5	B	6	C	4	D	3	E	3	F	1	G	0		
A	5																
B	6																
C	4																
D	3																
E	3																
F	1																
G	0																
b.	Explain the Knowledge-based agents with an algorithm and explain the TELL and ASK functions used for same.	10	CL2														
OR																	
6	a. Apply the A* search to find the solution path from a to z. Heuristics are with nodes, and cost is with edges. Write all steps as well as open and closed lists for full marks	10	CL3														
																	
b.	Considering Wumpus world example give the characteristics and simple knowledge base using propositional logic and Prove that Wumpus is in the room (1, 3).	10	CL3														
Module 4																	
7	a. List the drawbacks of Propositional Logic. Explain the Syntax and Semantics in First order Logic.	10	CL2														
	b. Define Universal and Existential Instantiation and give examples for both. Prove the following using Forward Chaining: "As per the law, it is a crime for an American to sell weapons to hostile nations. Country E, an enemy of America, has some missiles, and all the missiles were sold to it by Solan, who is an American citizen." Prove that "Solan is criminal."	10	CL3														
OR																	
8	a. Explain the following with respect to the first order logic: - Assertions and Queries in First order logic - The kinship domain - Numbers, sets and lists.	10	CL2														
	b. Considering the digital circuit example illustrate the seven steps in knowledge engineering process.	10	CL2														
Module 5																	
9	a. Explain Backward Chaining algorithm with an example.	10	CL2														

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	b. Define Classical Planning. With the blocks world example explain the same in detail.	10	CL2
OR			
10	a. Write appropriate quantifiers for the following (i) Some students read well (ii) Some students like some books (iii) Some students like all books (iv) All students like some books (v) All students like no books Explain the concept of Resolution in First Order Logic with appropriate procedure.	10	CL2
	b. Explain the two approaches to searching for a plan in detail.	10	CL2

Artificial Intelligence

Time: 3 hrs.

Max. Marks: 1

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

			Module – 1		M	L
Q.1	a.	What is Artificial Intelligence? Explain the four approaches of Artificial Intelligence in detail	10	L2		
	b.	What Artificial Intelligence can do today? Explain various application domains of Artificial Intelligence.	10	L2		
OR						
Q.2	a.	With the help of a block diagram, explain how agents interact with task environments.	05	L2		
	b.	Briefly explain the properties of task environments.	07	L2		
	c.	With a neat diagram explain the following : i) Utility based agents ii) Learning agents	08	L2		
Module – 2						
Q.3	a.	List and define the four phases of problem – solving process.	05	L3		
	b.	Explain how a search problem can be formally defined. Illustrate the same for the Vacuum world problem.	10	L2		
	c.	Write the state-space graph for the two-cell vacuum world problem.	05	L3		
OR						
Q.4	a.	Illustrate Breadth-first search and Depth-first search strategies with suitable example.	10	L3		
	b.	Explain the different metrics used to evaluate an algorithms performance.	06	L2		
	c.	List and explain the three kind of queues used in Best –first search method.	04	L2		
Module – 3						
Q.5	a.	Apply Greedy best-first search algorithm to the following graph and show the various stages in computing the solution tree. <pre> graph LR Start(()) --> A((A)) A --- 32 B((B)) A --- 40 C((C)) A --- 35 D((D)) B --- 19 E((E)) C --- 25 E C --- 17 F((F)) D --- 17 F E --- 10 H((H)) F --- 0 G((G)) G --- Goal(()) </pre> Fig. Q.5 (a)	10	L3		
	b.	Explain different methods of deriving heuristics.	10	L2		
OR						
Q.6	a.	Using PEAS description, represent the Wumpus World problem.	06	L3		
	b.	Briefly explain Knowledge Based Agents.	04	L2		
	c.	Explain in detail, syntax and semantics of propositional logic.	10	L3		

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Module – 4					
Q.7	a.	Explain in detail, syntax of first order logic.	10	L2	CO4
	b.	Explain various steps in the knowledge engineering process.	10	L2	CO4
OR					
Q.8	a.	Write Unification algorithm. Explain with suitable example.	10	L2	CO4
	b.	Explain simple forward – chaining algorithm with an example.	10	L2	CO4
Module – 5					
Q.9	a.	Explain Backward – chaining algorithm with an example.	10	L2	CO5
	b.	Illustrate the procedure to convert a sentence of first order logic to Conjunctive Normal Form (CNF)	10	L2	CO5
OR					
Q.10	a.	Illustrate categories and objects with suitable examples.	10	L2	CO5
	b.	Explain the following with respect to events: i) Time ii) Fluents and objects	10	L2	CO5



CBCS SCHEME

BCS515B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Artificial Intelligence

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.
2 M : Marks, L: Bloom's level, C: Course outcomes.

Module - 1				
Q.1	a.	Define the following : i) Intelligence ii) Artificial Intelligence iii) Agent iv) Rationality v) Logical reasoning.	5	L2 CO1
	b.	Examine the AI literature to discover whether the following tasks can currently be solved by computers. i) Playing a decent game of table tennis (ping-pong) ii) Discovering and proving new mathematical theorems iii) Giving competent legal advice in a specialized area of law iv) Performing a complex surgical operation.	8	L2 CO1
	c.	Implement a simple reflex agent for the vacuum environment. Run the environment with this agent for all possible initial dirt configurations and agent locations. Record the performance score for each configuration and the overall score.	7	L3 CO1
OR				
Q.2	a.	Is AI a science, or is it engineering or neither or both? Explain.	5	L2 CO1
	b.	Write pseudocode agent programs for the goal based and utility based agents.	8	L1 CO1
	c.	For each of the following activities give a PEAS description. i) Playing a tennis match ii) Performing a high jump iii) Bidding on an item in an auction.	7	L1 CO1
Module - 2				
Q.3	a.	Explain why problem formulation must follow goal transformation.	5	L1 CO1
	b.	Give complete problem formulation for each of the following choose a formulation that is precise enough to be implemented. i) Using only four colors, you have to color a planar graph in such a way that no two adjacent regions have the same color. ii) A 3-foot - foot - tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two-stackable, moveable, climbable 3-foot high crates.	8	L2 CO2
	c.	Prove each of the following statements or given counter example : i) Breadth - first search is a special case of uniform - cost search. ii) Uniform - cost search is a special case of A* search.	7	L2 CO2

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OR				
Q.4	a.	Define the following terms with example. i) State space ii) Search node iii) Transition model iv) Branching factor.	8	
	b.	Show that the 8-puzzle states are divided in to two disjoint sets, such that any state is reachable from any other state in the same set, while no state is reachable from any state in the other set. Devise a procedure to decide which set a given state is in and explain why this is useful for generating random state.	7	
	c.	Describe a state space in which iterative deepening search performs much worse than depth first search for example, $O(n^2) \forall n$.	5	
Module - 3				
Q.5	a.	Devise a state space in which A* using GRAPH-SEARCH returns a suboptimal solution with $h(n)$ function that is admissible but inconsistent.	7	
	b.	Which of the following are correct? i) $(A \vee B) \wedge (C \vee D) \vee E \wedge (A \vee B)$ ii) $(A \vee B) \wedge (C \vee D) \vee E \wedge (A \vee B) \wedge (\neg D \vee E)$ iii) $(A \vee B) \wedge (\neg A \vee B)$ is satisfiable iv) $(A \leftrightarrow B) \leftrightarrow C$ has the same number of models as $(A \leftrightarrow B)$	8	
	c.	Consider a vocabulary with only four propositions, A, B, C and D. How many models are there for the following sentences? i) $B \vee C$ ii) $\neg A \vee \neg B \vee \neg C \vee \neg D$ iii) $(A \supset B) \wedge A \wedge \neg B \wedge C \wedge D$	5	
OR				
Q.6	a.	Prove that if a heuristic is consistent, it must be admissible. Construct an admissible heuristic that is not consistent.	8	
	b.	Prove each of the following assertions : i) $\alpha \models \beta$ if and only if the sentence $(\alpha \leftrightarrow \beta)$ is valid ii) $\alpha \models \beta$ if and only if the sentence $\alpha \wedge \neg \beta$ is unsatisfiable.	7	
	c.	Prove, or find a counter example to each of the following assertions. i) If $\alpha \models (\beta \wedge \gamma)$ then $\alpha \models \beta$ and $\alpha \models \gamma$ ii) If $\alpha \models (\beta \vee \gamma)$ then $\alpha \models \beta$ and $\alpha \models \gamma$ (or) both	5	
Module - 4				
Q.7	a.	Which of the following are valid necessary true sentences? i) $\exists x (x = x) \Rightarrow (\forall y \exists x y = z)$ ii) $\forall x, P(x) \vee \neg p(x)$ iii) $\forall x, \text{smart}(x) \vee (x = x)$	7	
	b.	Prove that universal instantiation is sound that existential instantiation produces an inferentially equivalent knowledge base.	5	

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	c.	Write down logical representations for the following sentences, suitable for use with generalized modus ponens : i) Horses, cows and pigs are mammals ii) Bluebeard is Charlie's parent iii) Offspring and parent are inverse relations	8	L1 CO4
OR				
Q.8	a.	Consider a knowledge base containing just two sentence : $P(a)$ and $P(b)$. Does this knowledge base entail $\forall x P(x)$? Explain your answer in terms of models.	5	L2 CO4
	b.	Suppose a knowledge base contains just one sentence, $\exists x \text{AsHighAs}(x, \text{Everest})$ which of the following are legitimate results of applying existential instantiation? i) $\text{AsHighAs}(\text{Kilimanjaro}, \text{Everest})$ ii) $\text{AsHighAs}(\text{Kilimanjaro}, \text{Everest}) \wedge \text{AsHighAs}(\text{Beverly}, \text{Everest})$	8	L2 CO4
	c.	Explain how to write any 3-SAT problem of arbitrary size using a single first order definite clause and no more than 30 ground facts.	7	L2 CO4
Module - 5				
Q.9	a.	i) Give a backward chaining proof of the sentence? $7 \leq 3 + 9$. Show only the steps that leads to success ii) Give a forward chaining proof of the sentence $7 \leq 3 + 9$. Show only the steps that leads to success.	8	L1 CO5
	b.	Describe the differences, and similarities between problem solving and planning.	5	L2 CO5
	c.	Prove that backward search with PDDL problems is complete.	7	L1 CO5
OR				
Q.10	a.	The following prolog code defines a predicate P $P(x, [y,z])$ $P(x, [y,z]) :- P(x, z)$ i) Show proof trees and solutions for the queries $P(A, [2, 1, 3])$ and $P(x, [1, A, 3])$ ii) What standard list operation does P represent?	8	L1 CO5
	b.	Explain why dropping negative effects from every action schema in a planning problem results in a relaxed problems.	5	L2 CO5
	c.	Prove the following assertions about planning graphs : i) A literal that does not appear in the final level of the graph cannot be achieved. ii) The level cost of a literal in a serial graph is no greater than the actual cost of an optimal plan for achieving it.	7	L1 CO5

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